x_1	x_2	y
1	2	5
3	4	11

Use normal eq^n method & find θ

$$\theta = (X^T X)^{-1} X^T \vec{y}$$

Let $X = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $X^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$

$$\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

~~1x1 n 3x3~~

$4 + 16$

$$\begin{bmatrix} 10 & 14 \\ 14 & 20 \end{bmatrix} \Rightarrow X^T X$$

$$X^T \vec{y} = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \times \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

$5 + 33$

$$= \begin{matrix} 5 + 33 \\ 10 + 44 \end{matrix} = \begin{bmatrix} 38 \\ 54 \end{bmatrix}$$

$x^{(1)} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $y^{(1)} = 5$ | $x^{(2)} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $y^{(2)} = 11$

design matrix $= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = X$

$$\vec{y} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 10 & 14 \\ 14 & 20 \end{bmatrix}$$

$$(X^T X)^{-1} = \begin{bmatrix} 20 & -14 \\ -14 & 10 \end{bmatrix} \times \frac{1}{\cancel{28}}$$

$$\cancel{200 + 196} \rightarrow (20 \times 10) - (-14 \times -14)$$

$$200 - 112$$

$$\frac{1}{4} \cancel{\frac{1}{28}} \begin{bmatrix} 20 & -14 \\ -14 & 10 \end{bmatrix} \begin{bmatrix} 38 \\ 54 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\cancel{20 \times 38 - 14 \times 54} \quad \leftarrow 4$$

$$\cancel{-8 \times 58 \quad 54 \times 10}$$

$$\begin{bmatrix} 4 \\ 8 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \cancel{\frac{1}{2}}$$

$$\frac{1}{4} \times 4 = \theta_0 = 1$$

$$\frac{1}{4} \times 8 = \theta_1 = 2$$

$$\therefore \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$